

IGCSE · 4CH1

Chemistry — class test

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Total marks	40
Questions	40
Suggested time	≈ 40 min

Name: _____

Class: _____

Date: _____

Answer **all** questions. Show all working — marks are awarded for method as well as the final answer.

Chemistry — class test

Chemistry (4CH1) · IGCSE · total 40 marks · name: _____

Question 1 (1 mark)

(a)

[1]

Propanoic acid reacts with ethanol in an esterification reaction.

What is the displayed formula of the ester that is formed during this reaction?

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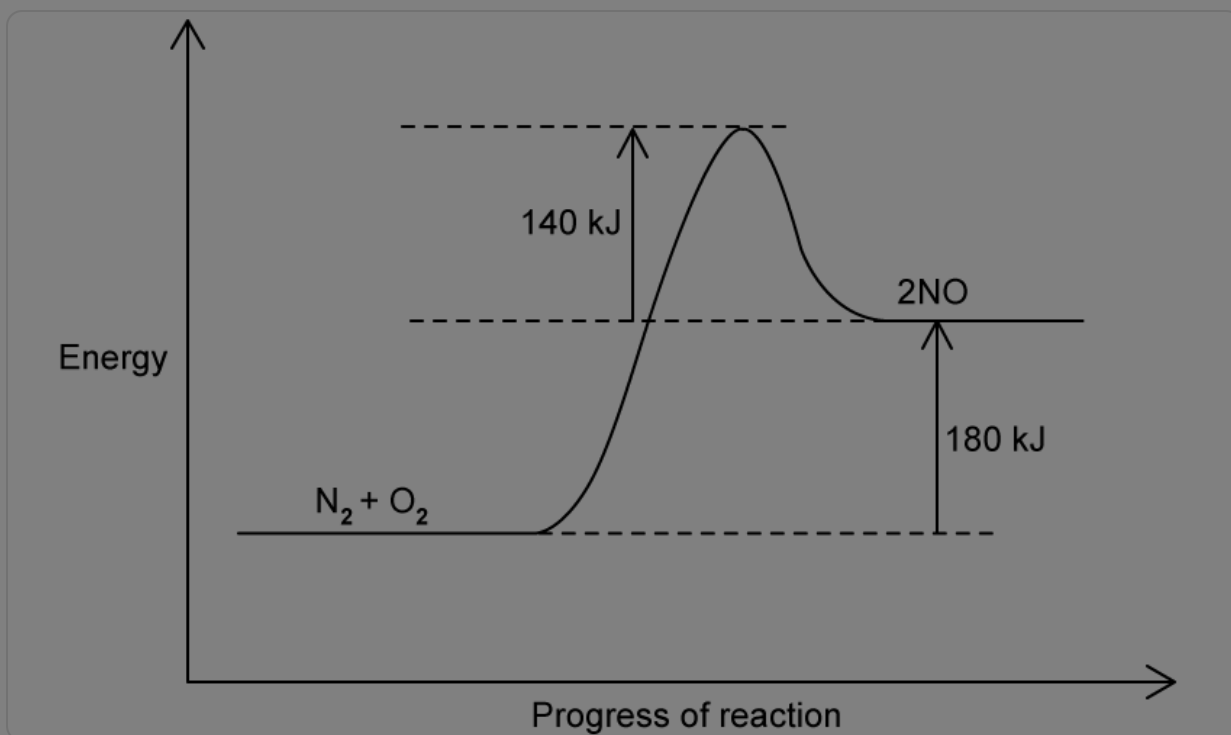
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Question 2 (1 mark)

(a)

[1]

The reaction profile for making nitrogen monoxide, NO, is shown below:



What is the activation energy for this reaction?

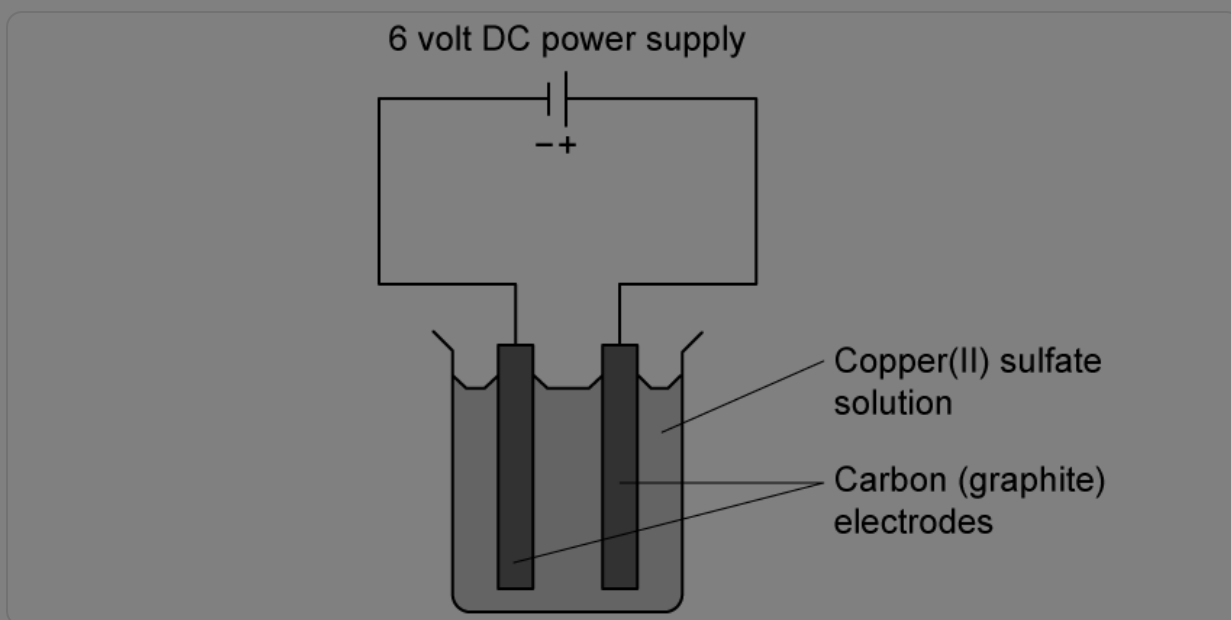
- A) +140 kJ
- B) +180 kJ
- C) +320 kJ
- D) -330 kJ

Question 3 (1 mark)

(a)

[1]

A student uses the apparatus to electrolyse an aqueous solution of copper(II) sulfate.



Which substances are formed at the negative electrode (cathode) and the positive electrode (anode)?

- A) Negative electrode: hydrogen Positive electrode: sulfur
- B) Negative electrode: copper Positive electrode: oxygen
- C) Negative electrode: sulfur Positive electrode: hydrogen
- D) Negative electrode: oxygen Positive electrode: copper

Question 4 (1 mark)

(a)

[1]

A company produces tin from an ore called cassiterite, which is mainly tin oxide.

What is the relative formula mass (M_r) of tin oxide, SnO_2 ?

[A_r of Sn = 119; O = 16]

- A) 135
- B) 151
- C) 270
- D) 1904

Question 5 (1 mark)

(a)

[1]

A student is carrying out a titration using phenolphthalein indicator.

What is the correct colour of this indicator in an acid and alkali?

- A) Colour in acid: pink Colour in alkali: colourless
- B) Colour in acid: colourless Colour in alkali: pink
- C) Colour in acid: red Colour in alkali: yellow
- D) Colour in acid: orange Colour in alkali: red

Question 6 (1 mark)

(a)

[1]

Which element in Group 7 is a solid at room temperature?

- A) fluorine
- B) chlorine
- C) bromine
- D) iodine

Question 7 (1 mark)

(a)

[1]

A compound is suspected of containing copper ions, Cu^{2+} and bromide ions, Br^- .

Which test would **not** help to positively identify copper ions or bromide ions?

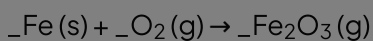
- A) Test: flame test Result: blue-green flame
- B) Test: add sodium hydroxide solution Result: blue precipitate forms
- C) Test: add acidified silver nitrate solution Result: cream precipitate forms
- D) Test: add sodium hydroxide solution Result: green precipitate forms

Question 8 (1 mark)

(a)

[1]

What numbers will balance the following equation?



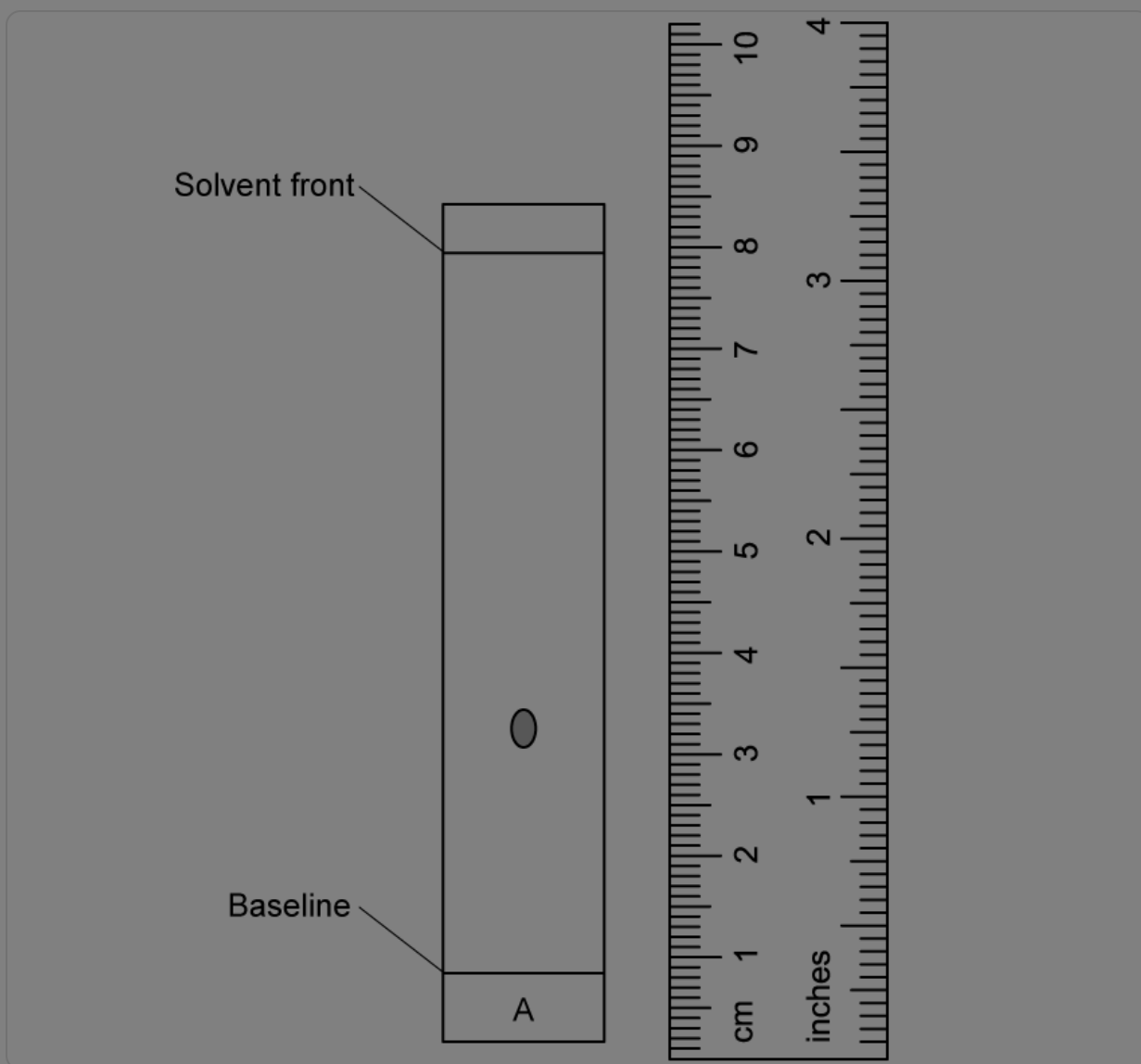
- A) 2,3,4
- B) 4,3,2
- C) 2,3,1
- D) 1,2,3

Question 9 (1 mark)

(a)

[1]

A student obtained the following chromatogram for substance A.



Determine the R_f value of substance A.

- A) 0.66
- B) 0.26
- C) 0.35
- D) 0.43

Question 10 (1 mark)

(a)

[1]

What type of reaction occurs when iron rusts?

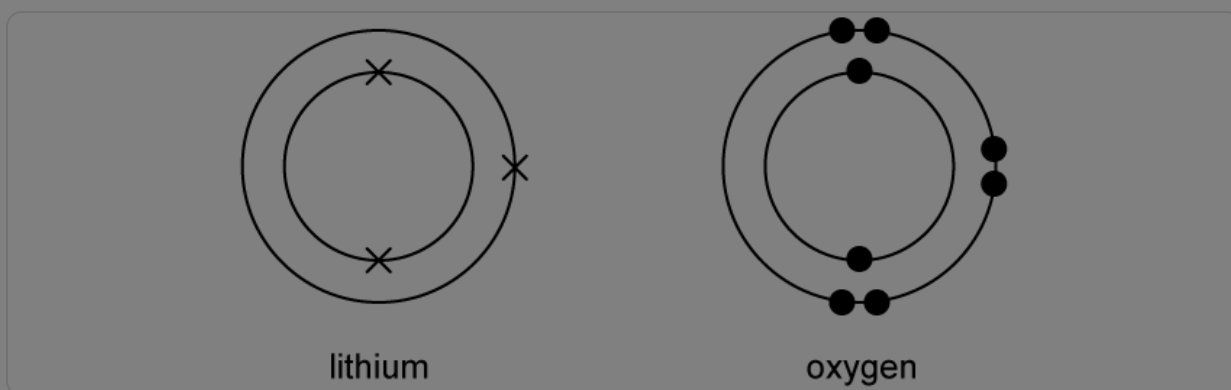
- A) Oxidation
- B) Neutralisation
- C) Combustion
- D) Displacement

Question 11 (1 mark)

(a)

[1]

The diagram shows the electronic configurations of an atom of lithium and an atom of oxygen.



Which statement is **not** correct about the changes in electronic configuration that occur when lithium and oxygen react to form lithium oxide, Li_2O ?

- A) Two lithium atoms each lose one electron
- B) One oxygen atom gains 2 electrons
- C) A lithium ion has an electronic configuration of 2
- D) An oxide ion has an electronic configuration of 2.6

Question 12 (1 mark)

(a)

[1]

Carbonic acid is found in fizzy drinks and has the formula, H_2CO_3 .

Which ion causes the drink to be acidic?

- A) OH^-
- B) CO_3^{2-}
- C) H^+
- D) O^{2-}

Question 13 (1 mark)

(a)

[1]

Carbon dioxide contains covalent bonds.

The covalent bonds in a molecule of carbon dioxide can be represented by a dot-and-cross diagram.

Which is the correct dot-and-cross diagram for carbon dioxide?

.....

.....

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Question 14 (1 mark)

(a)

[1]

Which of the following reactions is an example of an addition reaction?

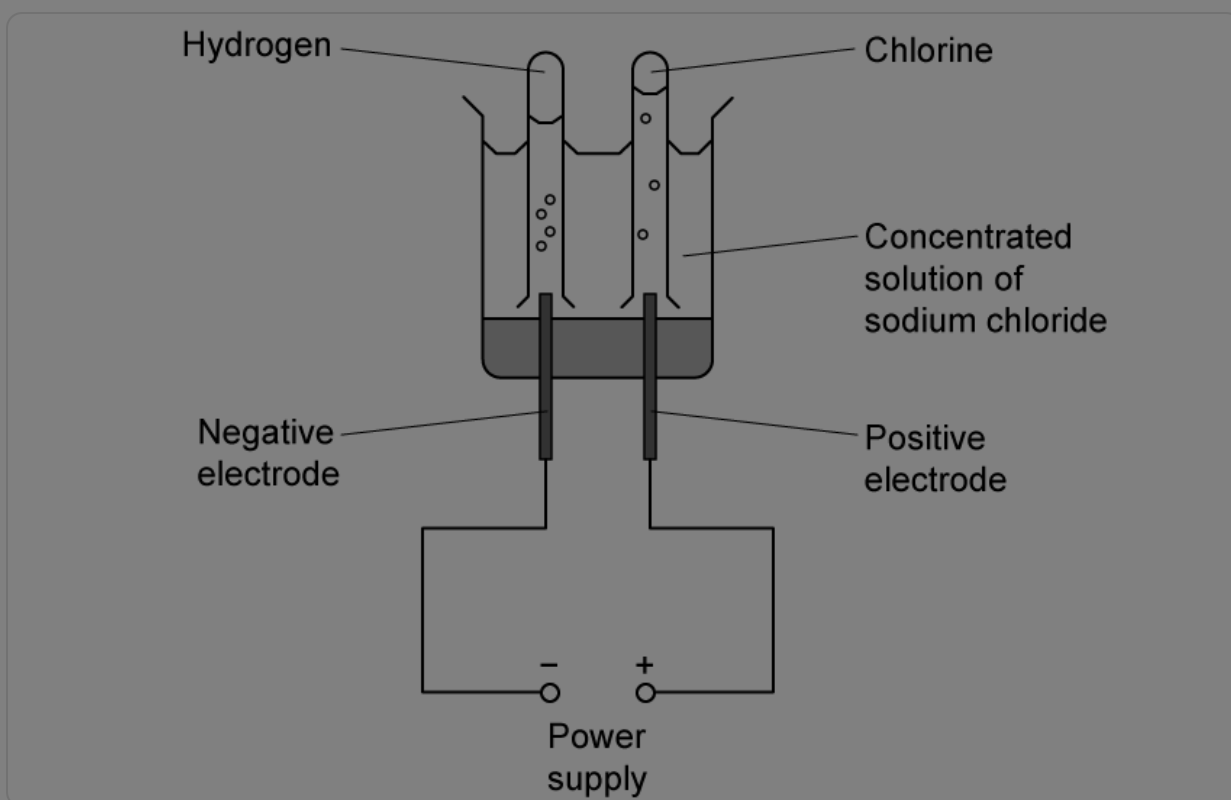
- A) $\text{CH}_4 + \frac{1}{2}\text{O}_2 \rightarrow \text{CO} + 2\text{H}_2\text{O}$
- B) $\text{C}_2\text{H}_4 + \text{Br}_2 \rightarrow \text{C}_2\text{H}_4\text{Br}_2$
- C) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- D) $\text{CH}_4 + \text{Br}_2 \rightarrow \text{CH}_3\text{Br} + \text{HBr}$

Question 15 (1 mark)

(a)

[1]

Hydrogen gas and chlorine gas can be prepared in a laboratory by electrolysis of a concentrated solution of sodium chloride.



Which is the ionic half-equation that represents the formation of chlorine gas at the positive electrode?

- A) $\text{Cl}^- \rightarrow \text{Cl} + \text{e}^-$
- B) $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
- C) $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$
- D) $2\text{Cl}^- + 2\text{e}^- \rightarrow \text{Cl}_2$

Question 16 (1 mark)

(a)

[1]

Electrolysis can be used to separate molten lead bromide.

Which statement is true for what happens at the negative electrode?

☐	A	Lead atoms are oxidised
☐	B	Lead atoms are reduced
☐	C	Lead ions are oxidised
☐	D	Lead ions are reduced

- A) Lead atoms are oxidized
- B) Lead atoms are reduced
- C) Lead ions are oxidized
- D) Lead ions are reduced

Question 17 (1 mark)

(a)

[1]

Which of the following reactions does **not** involve the oxidation of ethanol?

- A) Production of ethanol from the fermentation of glucose
- B) Heating ethanol with potassium dichromate(VI) in dilute sulfuric acid
- C) Reaction of ethanol with oxygen in the air
- D) Formation of carbon dioxide and water from the complete combustion of ethanol

Question 18 (1 mark)

(a)

[1]

Which equation does **not** show the correct reaction of an acid?

- A) copper oxide + hydrochloric acid → copper chloride + water
- B) calcium carbonate + nitric acid → calcium nitrate + carbon dioxide
- C) potassium hydroxide + sulfuric acid → potassium sulfate + water
- D) zinc + sulfuric acid → zinc sulfate + hydrogen

Question 19 (1 mark)

(a)

[1]

Fractional distillation is the process by which crude oil is separated into simpler, more useful fractions.

Which of the following statements about the process of fractional distillation is correct?

- A) The fractionating column is cooler at the bottom than at the top
- B) The fractions collected at the top of the fractionating column have a high boiling point
- C) The molecules in each fraction have a similar boiling point and carbon chain length
- D) The fractions collected at the bottom of the fractionating column have a low viscosity

Question 20 (1 mark)

(a)

[1]

Ethanoic acid is a member of the homologous series of carboxylic acids.

Which is the functional group of carboxylic acids?

.....

.....

.....

Question 21 (1 mark)

(a)

[1]

Which of the statements describes ionic bonding?

- A) Electrostatic attraction between atoms
- B) Electrostatic attraction between positively charged particles and delocalized electrons
- C) Electrostatic attraction between oppositely charged ions
- D) Electrostatic attraction between the nuclei of two atoms and a shared pair of electrons

Question 22 (1 mark)

(a)

[1]

A student added some sulfuric acid gradually to a solution of ammonia until it was in excess.

What happens to the pH during this reaction?

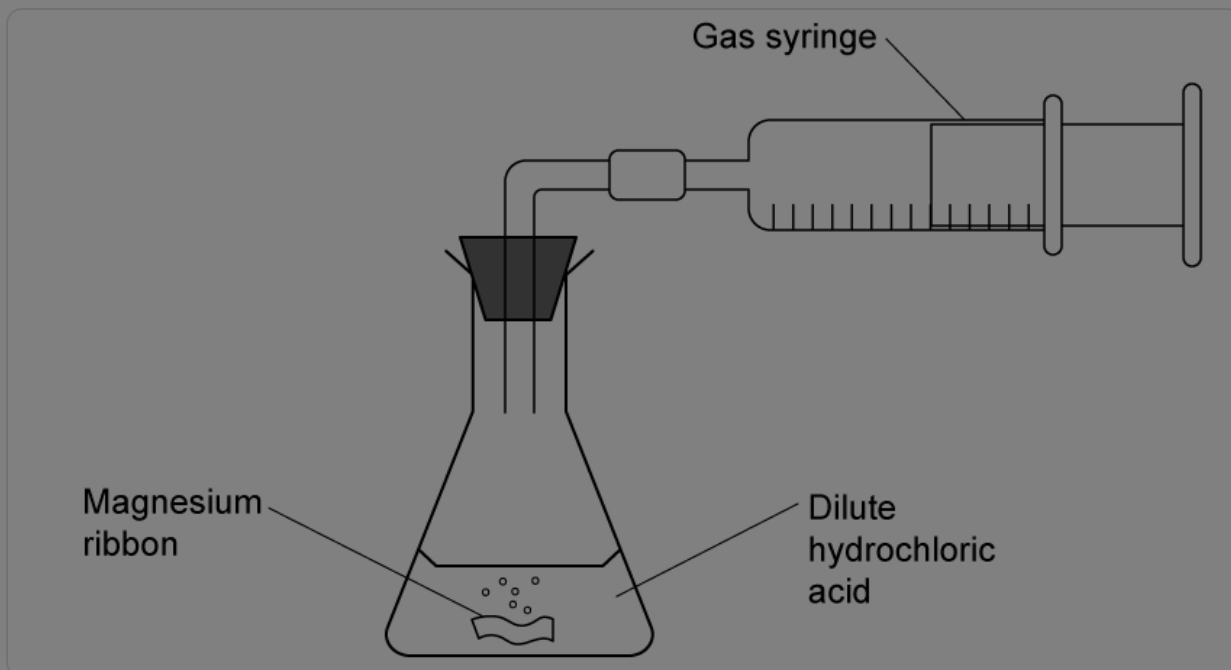
- A) pH of ammonia at start: 11 pH after excess sulfuric acid: 7
- B) pH of ammonia at start: 7 pH after excess sulfuric acid: 13
- C) pH of ammonia at start: 11 pH after excess sulfuric acid: 2
- D) pH of ammonia at start: 3 pH after excess sulfuric acid: 8

Question 23 (1 mark)

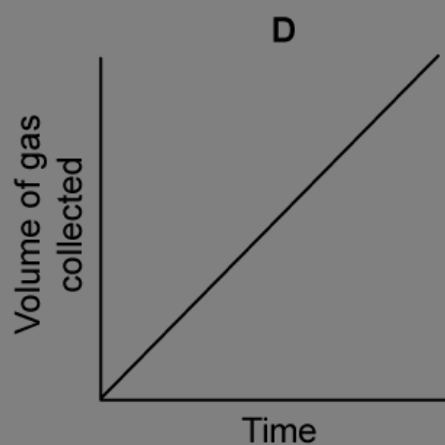
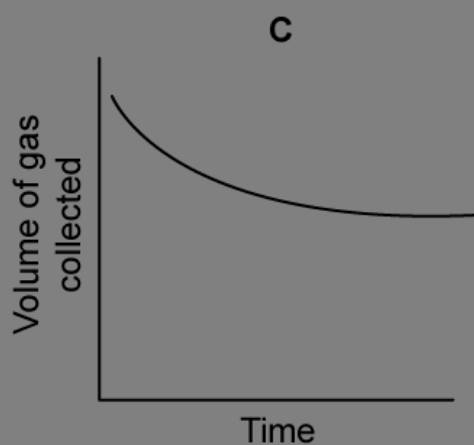
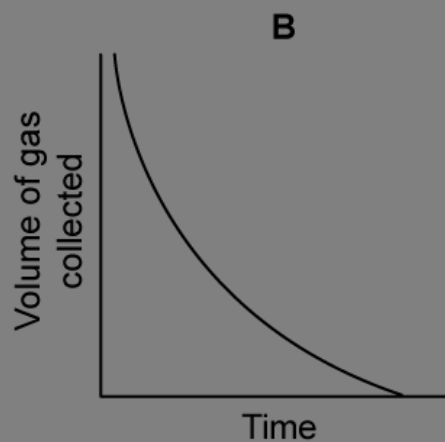
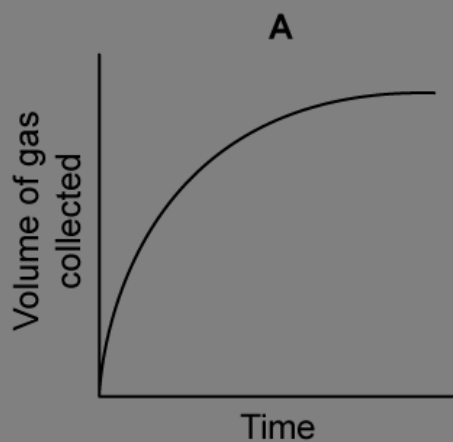
(a)

[1]

A student investigates the rate of reaction between an **excess** of hydrochloric acid and a limiting mass of magnesium using the apparatus shown:



Which of the graphs could represent her results?



Question 24 (1 mark)

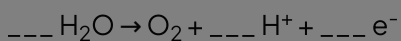
(a)

[1]

When copper(II) sulfate undergoes electrolysis, it produces oxygen gas at the positive electrode.

The unbalanced half equation below represents the reaction that occurs at the positive electrode.

Which numbers will correctly balance the half equation?



☐	A	2,2,2
☐	B	2,4,2
☐	C	2,4,4
☐	D	4,4,2

A) 2,2,2

B) 2,4,2

C) 2,4,4

D) 4,4,2

Question 25 (1 mark)

(a)

[1]

Sodium oxide reacts with water to form sodium hydroxide.

A scientist plans to measure the temperature change of the reaction using calorimetry in a lab.

This is the scientist's basic method:

- use 50 g of water
- record the temperature of the water
- add sodium hydroxide
- stir
- record the maximum temperature reached

Which apparatus list is the most appropriate for this experiment?

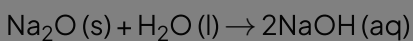
- A) Electronic balance, measuring cylinder, glass beaker, thermometer
- B) Measuring cylinder, glass beaker, polystyrene cup, thermometer
- C) Electronic balance, glass beaker, polystyrene cup, thermometer
- D) Measuring cylinder, glass beaker, glass rod, thermometer

Question 26 (1 mark)

(a)

[1]

Another student carries out the experiment to find the heat energy released in the reaction of sodium oxide with water:



They found that 10,800 J of heat energy was released when 0.060 mol of sodium oxide reacts.

What is the value of ΔH , in kJ/mol, for the reaction of sodium oxide with water?

- A) 0.0056 kJ/mol
- B) 180 kJ/mol
- C) 1,800 kJ/mol
- D) 8,000 kJ/mol

Question 27 (1 mark)

(a)

[1]

The diagram shows part of the periodic table, with elements represented by the letters M, Q, R and T. The letters in the diagram represent elements but are not their chemical symbols.

Use the Periodic Table to help you answer this question.

																M	
	Q											R					
T																	

Which of these statements is true?

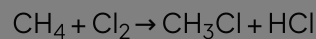
- A) T and Q form acidic oxides
- B) M conducts electricity
- C) M is a non-metal element
- D) R is a gaseous non-metal element

Question 28 (1 mark)

(a)

[1]

Chlorine reacts with methane in sunlight.



The diagram below shows the displayed formulae for the reaction of chlorine with methane.



The table below shows the bond energies and the overall energy change in the reaction.

	C-H	Cl-Cl	C-Cl	H-Cl	Overall energy change
Energy in kJ/mol	413	243	X	432	-122

What is the value of X?

- A) 67
- B) 134
- C) 346
- D) 542

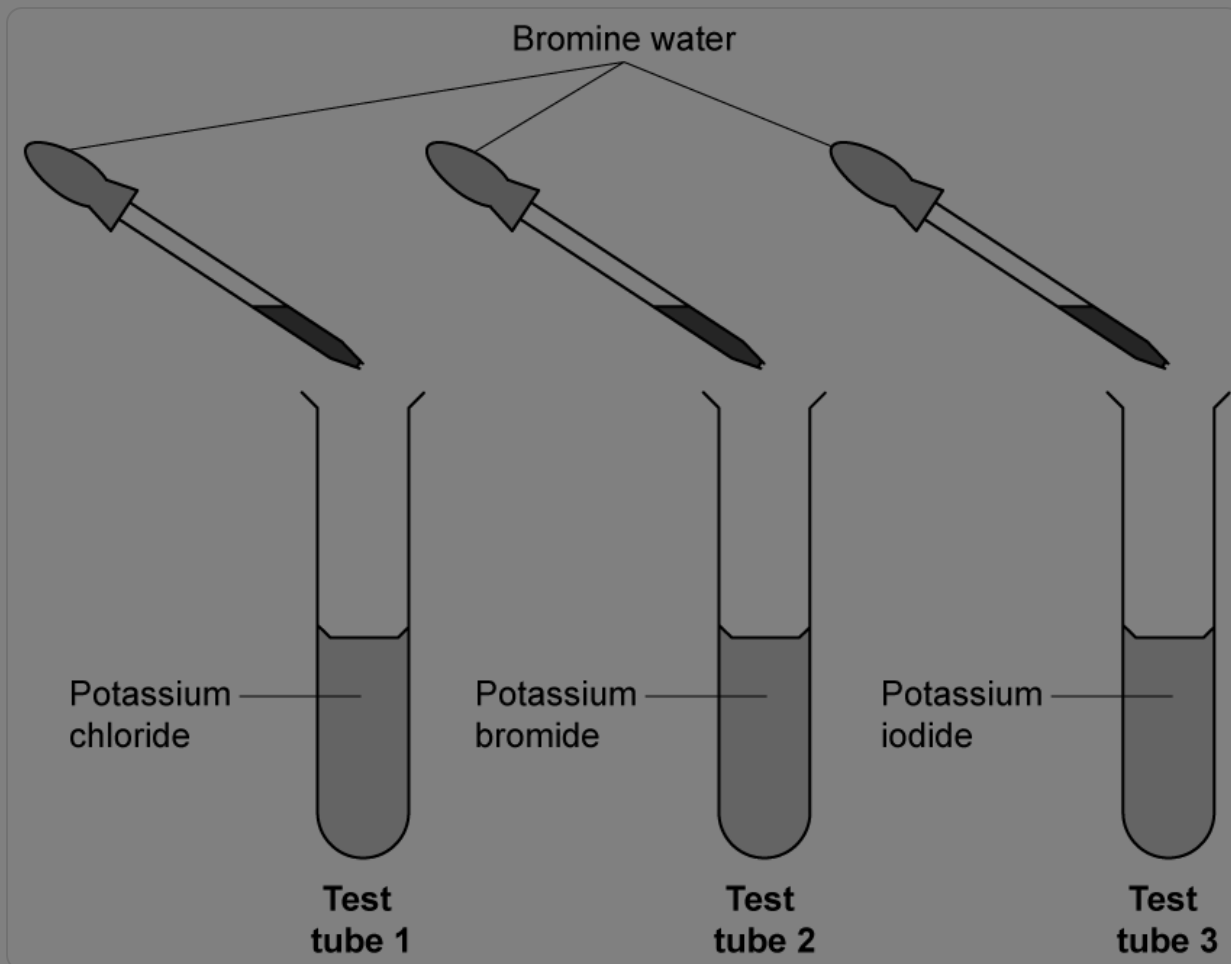
Question 29 (1 mark)

(a)

[1]

Bromine water is an aqueous solution of bromine which is orange in colour.

What would be seen if bromine water was added to the test tubes in the diagram?



	Test tube 1	Test tube 2	Test tube 3
☐ A	solution stays orange	solution turns brown	solution turns brown
☐ B	solution turns pale green	solution stays orange	solution turns colourless
☐ C	solution stays orange	solution stays orange	solution turns brown
☐ D	solution turns colourless	solution turns colourless	solution turns brown

Question 30 (1 mark)

(a)

[1]

Scientists test a solution of compound X. The table shows their results.

Test	Result
Acidified silver nitrate solution	White precipitate
Add sodium hydroxide solution	Green precipitate

Which **two** ions are present in compound X?

☐	A	Fe^{3+} and Cl^-
☐	B	Fe^{3+} and SO_4^{2-}
☐	C	Fe^{2+} and Cl^-
☐	D	Fe^{2+} and SO_4^{2-}

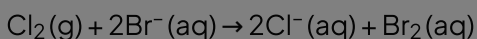
- A) Fe^{3+} and Cl^-
B) Fe^{3+} and SO_4^{2-}
C) Fe^{2+} and Cl^-
D) Fe^{2+} and SO_4^{2-}

Question 31 (1 mark)

(a)

[1]

The ionic equation for the reaction between chlorine and sodium bromide is:



This reaction occurs because chlorine is more reactive than bromine.

Which of the statements explaining why chlorine is more reactive than bromine is **not** true?

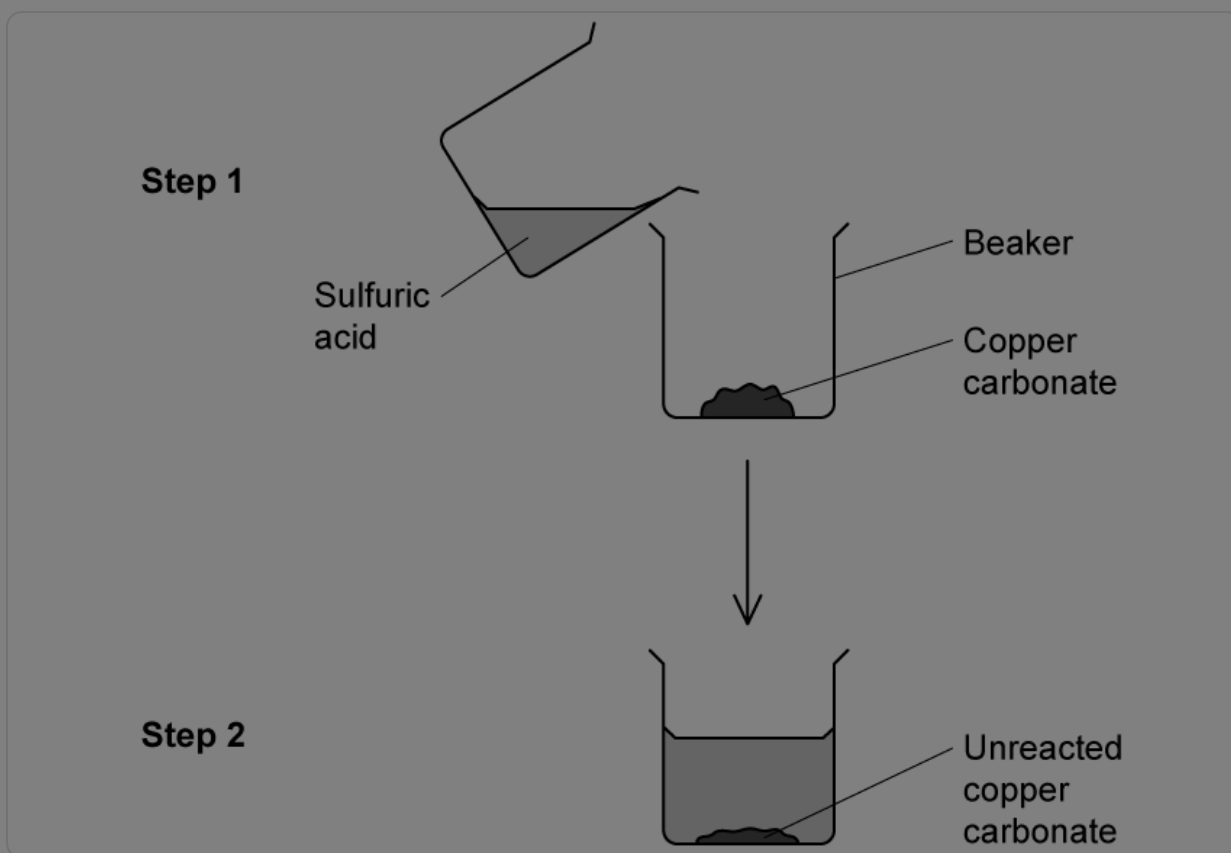
- A) chlorine has smaller atoms than bromine
B) chlorine atoms gain electrons more easily
C) chlorine atoms lose electrons more easily
D) chlorine atoms have fewer electron shells than bromine atoms

Question 32 (1 mark)

(a)

[1]

A student making copper sulfate crystals used the method below.



Unreacted copper carbonate was left over as it had been added in excess.

What is the reason for adding it in excess and what would step 3 be of this method?

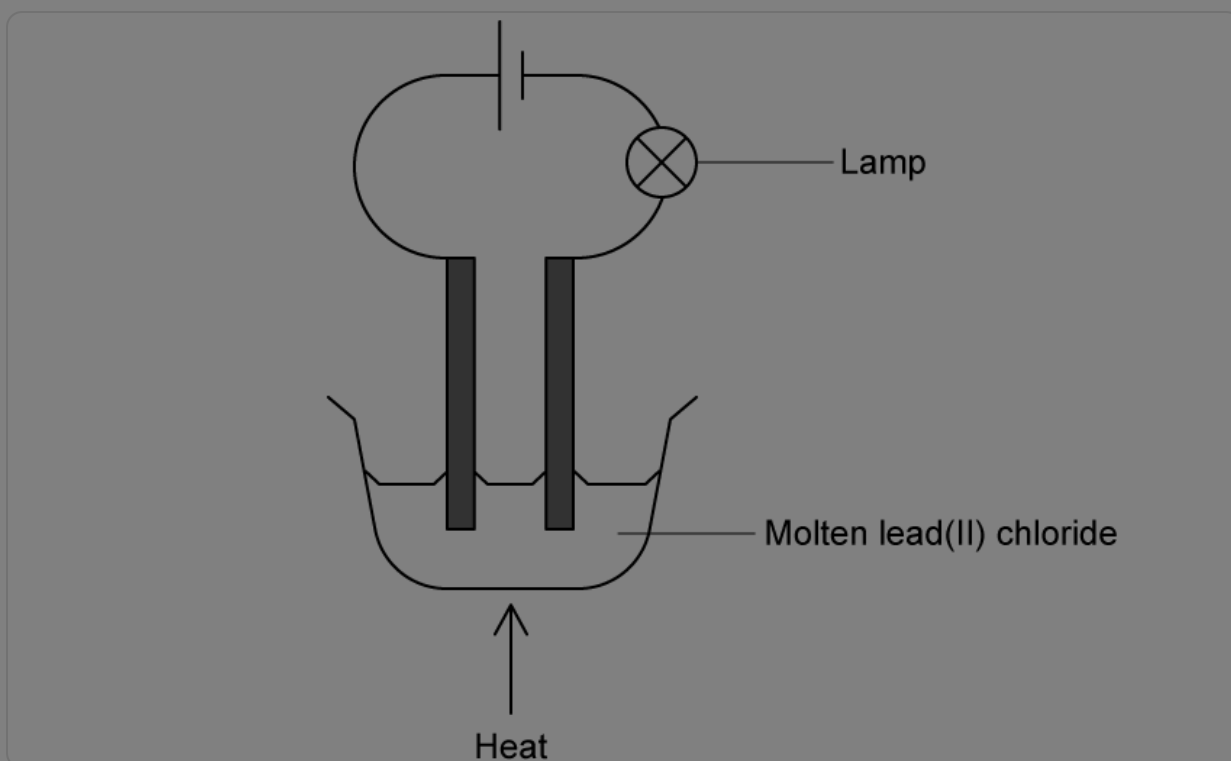
- A) Reason: to produce a greater amount of salt crystals Step 3: filtration
- B) Reason: to improve the colour intensity of the crystals Step 3: crystallisation
- C) Reason: to ensure all the acid reacts Step 3: filtration
- D) Reason: to increase the rate of reaction Step 3: evaporation

Question 33 (1 mark)

(a)

[1]

A pure sample of molten lead(II) chloride was electrolysed.



Which is the correct statement about this electrolysis?

- A) The lamp would stop working soon after the heat was removed
- B) A silvery liquid would be seen at the positive electrode
- C) Chlorine is formed at the negative electrode
- D) Hydrogen gas is formed at the negative electrode

Question 34 (1 mark)

(a)

[1]

The actual molar enthalpy change, ΔH , for the reaction of sodium oxide with water, is -238 kJ/mol .

Experimental results from calorimetry in labs give less exothermic results.

Which of the following does not explain this difference?

- A) The reaction may have been incomplete
- B) Heat energy also transfers to the surroundings
- C) The calorimetry equipment also has a specific heat capacity, and so affects the final temperature reading
- D) The student measured the initial temperature of the water as lower than the actual value.

Question 35 (1 mark)

(a)

[1]

Butanol is a member of the homologous series of alcohols.

Which of the following is **not** a formula of butanol?

- A) $\text{C}_4\text{H}_{10}\text{O}$
- B) $\text{C}_4\text{H}_{10}\text{OH}$
- C) $\text{C}_4\text{H}_9\text{OH}$
- D) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

Question 36 (1 mark)

(a)

[1]

A neutralisation reaction occurs between ammonia and sulfuric acid.

How does the ammonia act in this reaction?

- A) As a proton donor
- B) As an electron acceptor
- C) As a proton acceptor
- D) As a neutron donor

Question 37 (1 mark)

(a)

[1]

Which statement describes metallic bonding?

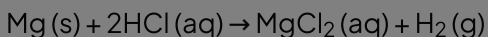
- A) Electrostatic attraction between atoms
- B) Electrostatic attraction between positively charged particles and delocalised electrons
- C) Electrostatic attraction between oppositely charged ions
- D) Electrostatic attraction between the nuclei of two atoms and a shared pair of electrons

Question 38 (1 mark)

(a)

[1]

Magnesium metal reacts with an excess of hydrochloric acid solution to form magnesium chloride and hydrogen:



Which of the following will **not** increase the rate of this reaction?

- A) Increase the temperature of the acid
- B) Increase the surface area of the pieces of magnesium
- C) Increase the concentration of the hydrochloric acid
- D) Increase the volume of hydrochloric acid solution used

Question 39 (1 mark)

(a)

[1]

Which of these statements about catalysts is **not** correct?

- A) The mass of the catalyst at the start of the reaction is the same at the end of the reaction
- B) Catalysts increase the rate of reaction but do not increase the yield
- C) Catalysts are not chemically changed during the course of the reaction
- D) Catalysts provide an alternative reaction pathway which has a lower activation energy

Question 40 (1 mark)

(a)

[1]

What is the relative formula mass of hydrated magnesium carbonate, $\text{MgCO}_3 \cdot 2\text{H}_2\text{O}$?

[A_r of Mg = 24; H = 1; O = 16; C = 12]

- A) 84
- B) 102
- C) 112
- D) 120